

Python Functions: 50 Practice Questions

Basic Functions (1-10)

1. Write a function that prints 'Hello, Python!'.
2. Write a function that accepts a name as a parameter and prints a greeting.
3. Write a function that takes two numbers and returns their sum.
4. Write a function that takes three numbers and returns the largest number.
5. Write a function that accepts a number and returns whether it is even or odd.
6. Write a function that calculates the square of a number.
7. Write a function that accepts a string and returns its length without using len().
8. Write a function that takes a list and returns the smallest element.
9. Write a function that accepts a number and returns its factorial.
10. Write a function that takes a year as input and returns whether it is a leap year or not.

Intermediate Functions (11-20)

11. Write a function that counts the number of vowels in a string.
12. Write a function that checks whether a string is a palindrome.
13. Write a function that returns all prime numbers between two given numbers.
14. Write a function that calculates the sum of all elements in a list.
15. Write a function that returns the second largest number from a list.
16. Write a function that accepts a sentence and returns the word with the maximum length.
17. Write a function that counts the frequency of each character in a string.
18. Write a function that removes duplicate elements from a list without using set().
19. Write a function that checks whether two strings are anagrams.
20. Write a function that returns the Fibonacci series up to n terms.

Parameters & Arguments (21-25)

21. Write a function using default arguments to calculate simple interest.
22. Write a function using keyword arguments to create a student profile.
23. Write a function using *args that returns the sum of any number of values.
24. Write a function using **kwargs that displays all employee details.
25. Write a function that combines *args and **kwargs in the same program.

Recursion (26-30)

26. Write a recursive function to calculate factorial.
27. Write a recursive function to calculate the nth Fibonacci number.

28. Write a recursive function to find the sum of digits of a number.
29. Write a recursive function to reverse a string.
30. Write a recursive function to calculate $\text{base}^{\text{power}}$.

Advanced Functions (31-40)

31. Write a function that acts as a custom version of `max()`.
32. Write a function that returns another function.
33. Create a closure that remembers previous values.
34. Write a decorator that prints messages before and after function execution.
35. Write a decorator that measures execution time of a function.
36. Write a function that accepts another function as an argument.
37. Implement a custom version of Python's `map()` function.
38. Implement a custom version of Python's `filter()` function.
39. Write a function that caches previously calculated results (Memoization).
40. Build a mini calculator using functions.

Interview & Conceptual Questions (41-50)

41. Can a function return multiple values? Demonstrate with an example.
42. What is the difference between `print()` and `return` inside a function?
43. What happens if a function does not have a `return` statement?
44. What is the difference between local and global variables inside functions?
45. What is variable shadowing in functions?
46. What is the difference between `def fun(a=[])` and `def fun(a=None)`?
47. Explain pass-by-value vs pass-by-reference behavior in Python functions.
48. What is the difference between `*args` and `**kwargs`?
49. What are lambda functions, and when should they be used?
50. When should you use recursion instead of loops?